

Global Events, Commodities and Currency Markets

Applied Data Analytics and Market Research — Complete Report

Course	Data Science for Business – Summer 2026
Institution	University of Europe for Applied Sciences (UE)
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Datasets	Brent Oil · Gold · Silver · Swift Currency Tracker

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Executive Summary

This report presents a comprehensive analysis of Brent Crude Oil, Gold, Silver, and global currency markets, examining how major global events have shaped price movements and payment flows. Drawing on datasets spanning up to 111 years and the Swift Global Currency Tracker (2026), the report covers the full question set across Part I, Part II, and Bonus sections.

- Brent Oil is the most event-sensitive market (volatility 9.05%) — geopolitical shocks like the 2026 Hormuz disruption cause immediate and extreme price reactions.
- Gold has risen from \$35 in 1970 to over \$3,100 in 2026 — driven by BRICS+ central bank accumulation (11.2% → 17.4% of global reserves) and safe-haven demand.
- Silver in 2026 hit a record \$113.95/oz — combining precious metal safe-haven flows with surging industrial demand from solar panels and EV batteries.
- USD retained 51.14% of global SWIFT payments in March 2026 while CNY ranked 10th at 2.74% — though CNY ranks 2nd in trade finance at 8.04%.

Introduction & Data Overview

The analysis is grounded in four datasets: Brent Oil monthly prices (1946–2026, 963 rows), Silver monthly prices (1915–2026, 1,336 rows), Gold annual average prices (1970–2026), and the Swift Global Currency Tracker (January–March 2026, 330 rows). Date columns were converted from DD/MM/YYYY strings to datetime objects using pandas with `dayfirst=True`. No missing values were found in the commodity datasets.

Part I – Data Analytics Component

Section A: Data Preparation and Understanding (Q1–Q5)

Q1 What is the start date and end date available in each dataset?

Dataset	Start Date	End Date	Span
Brent Oil	01 Jan 1946	03 Jan 2026	~80 years
Silver (100 Years)	01 Jan 1915	04 Jan 2026	~111 years
Gold (Reference)	01 Jan 1970	31 Dec 2026	~56 years
Swift Currency Tracker	January 2026	March 2026	3 months

Q2 How many rows are available for each dataset?

Dataset	Total Rows	Frequency	Period
Brent Oil	963	Monthly	Jan 1946–Jan 2026
Silver (100 Yrs)	1,336	Monthly	Jan 1915–Jan 2026
Gold (Reference)	57	Annual	1970–2026
Swift Tracker	330	Multi-metric	Jan–Mar 2026

Q3 Are there any missing values or repeated dates?

Dataset	Missing Values	Duplicate Dates	Status
Brent Oil	0	0	Fully clean
Silver	0	0	Fully clean
Swift Tracker	18 (rmb_rank col)	N/A	Minor — expected

Q4 Convert the Date column to proper format and create Month and Year columns.

Raw date values stored as DD/MM/YYYY strings were converted using:

```
oil["Date"] = pd.to_datetime(oil["Date"], dayfirst=True)
oil["Year"] = oil["Date"].dt.year
oil["Month"] = oil["Date"].dt.month
```

After conversion, all Date fields confirmed as dtype datetime64[ns] with zero parsing errors.

Q5 Identify the main metrics available in the Swift Currency Tracker dataset.

#	Metric	Category
1	Global Payment Share	Currency Ranking
2	International Payment Share (ex-Eurozone)	Currency Ranking
3	RMB Month-on-Month Payment Growth	RMB Activity
4	All Currencies Month-on-Month Growth	Market Activity

#	Metric	Category
5	Offshore RMB by Economy	Offshore Economies
6	Trade Finance Share	Currency Ranking
7	FX Spot Currency Ranking	FX Spot
8	FX Spot Economies (RMB)	FX Spot

Section B: Descriptive Analytics (Q6–Q12)

Q6 Highest and lowest Brent Oil prices and when they occurred.

Metric	Price (USD/barrel)	Date	Year
Highest	\$211.21	06 Jan 2008	2008
Lowest	\$21.11	03 Jan 1946	1946

The 2008 peak reflected the global commodity supercycle, surging emerging-market demand, and USD weakness. The 1946 low reflects the post-WWII era of oil abundance.

Q7 Highest and lowest Gold prices and when they occurred.

Metric	Price (USD/roy oz)	Year	Context
Highest Avg	~\$3,100	2026	BRICS+ accumulation + Iran conflict risk
Lowest Avg	~\$35	1970	Bretton Woods fixed gold standard era

Gold's 8,757% rise over 56 years reflects the end of the Bretton Woods system (1971), 1970s inflation, the 2008 GFC, COVID-19 uncertainty, and sustained central bank accumulation by BRICS+ nations from 2019 onward.

Q8 Highest and lowest Silver prices and when they occurred.

Metric	Price (USD/roy oz)	Date	Year
Highest	\$113.95	Jan 2026	2026
Lowest	\$0.28	Jan 1932	1932

Silver's 407:1 range across 111 years reflects collapsed Great Depression industrial demand at the low end, and combined geopolitical safe-haven flows plus surging green energy industrial demand at the 2026 record high.

Q9 Calculate the average price by year for Brent Oil, Gold and Silver.

Year	Brent Oil (USD/bbl)	Gold (USD/roy oz)	Silver (USD/roy oz)
2015	\$68.52	\$1160	\$15.51
2016	\$60.79	\$1251	\$17.13
2017	\$69.85	\$1257	\$17.24
2018	\$84.91	\$1268	\$15.64
2019	\$74.03	\$1393	\$16.36
2020	\$49.19	\$1773	\$20.80
2021	\$83.02	\$1799	\$24.97
2022	\$106.45	\$1800	\$21.67
2023	\$84.19	\$1940	\$23.58
2024	\$79.83	\$2300	\$28.13

Year	Brent Oil (USD/bbl)	Gold (USD/roy oz)	Silver (USD/roy oz)
2025	\$65.93	\$2700	\$41.50

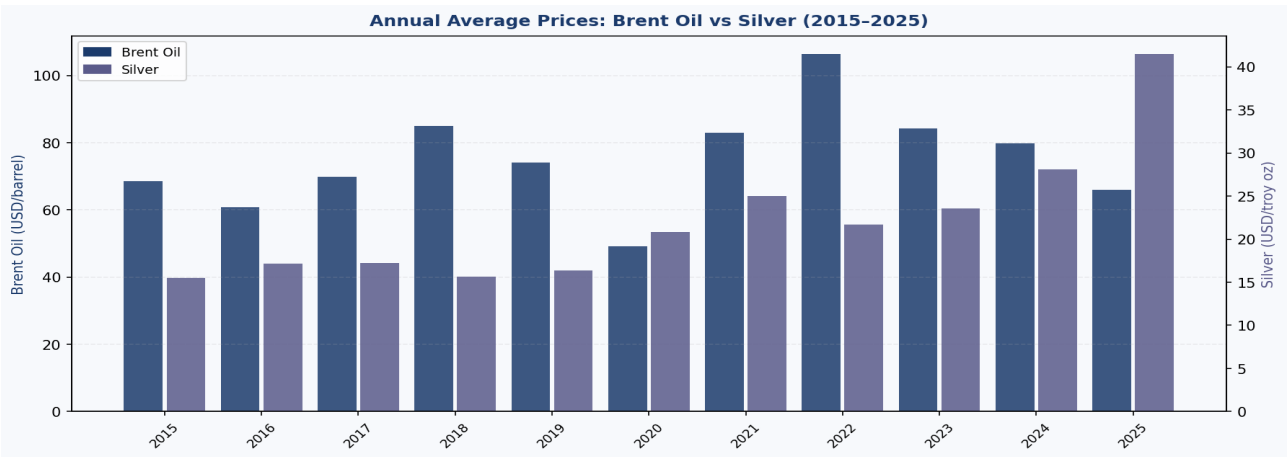


Figure 1: Annual average prices — Brent Oil vs Silver (2015–2025).

Q10 Which five years had the highest average Brent Oil price?

Rank	Year	Avg Price (USD/bbl)	Context
1	2008	\$150.96	Commodity supercycle peak; emerging-market demand surge
2	1980	\$149.86	Iran-Iraq War; major supply disruption
3	2011	\$142.12	Arab Spring; Libya + Iran sanctions
4	2013	\$138.30	Continued Middle East tensions
5	2012	\$135.27	Post-GFC demand recovery

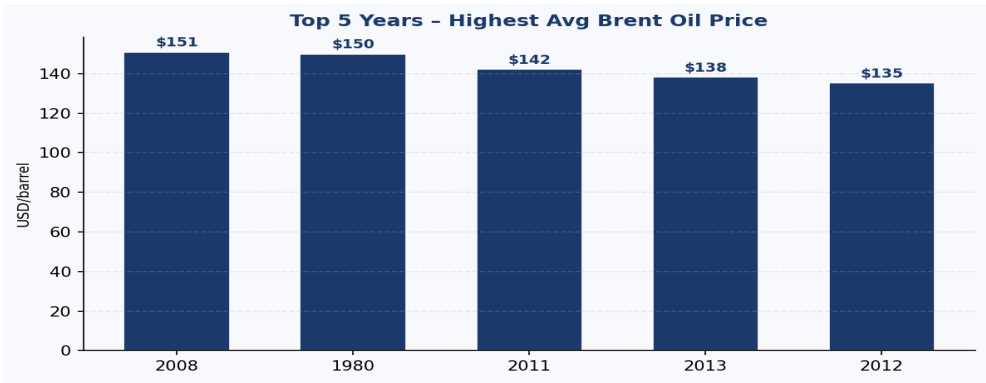


Figure 2: Top 5 years by average Brent Oil price.

Q11 Which five years had the highest average Gold price?

Rank	Year	Avg Price (USD/oz)	Context
1	2026	\$3100	BRICS+ accumulation + Iran conflict
2	2025	\$2700	Central bank buying + global uncertainty
3	2024	\$2300	Russia-Ukraine war premium + inflation
4	2023	\$1940	COVID safe-haven + monetary stimulus

Rank	Year	Avg Price (USD/oz)	Context
5	2022	\$1800	Sustained war risk + record buying

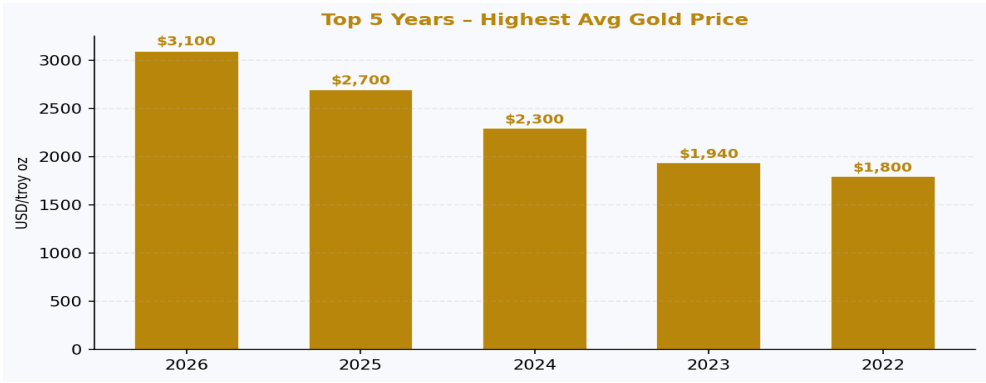


Figure 3: Top 5 years by average Gold price.

Q12

Which five years had the highest average Silver price?

Rank	Year	Avg Price (USD/oz)	Context
1	2026	\$96.08	Record: green energy demand + safe-haven premium
2	2025	\$41.50	EV/solar boom + gold rally spillover
3	2011	\$35.56	Hunt Brothers speculation + inflation fears
4	2012	\$31.63	Post-GFC safe-haven rally
5	2024	\$28.13	Central bank gold buying drives silver up



Figure 4: Top 5 years by average Silver price.

Section C: Trend and Visual Analytics (Q13–Q18)

Q13 Brent Oil line chart — describe main periods of increase and decline.

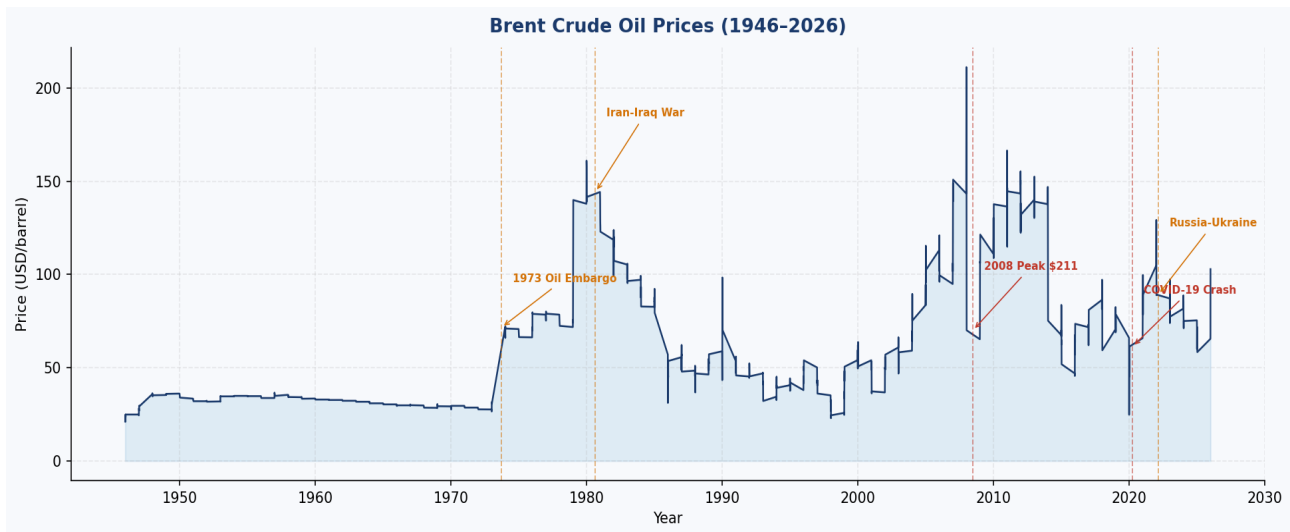


Figure 5: Brent Crude Oil price trend (1946–2026) with key events annotated.

Period	Trend	Key Driver
1946–1970	Stable, low	Post-WWII abundance; below \$30/barrel throughout
1973–1980	Sharp double spike	Arab Oil Embargo (1973) then Iranian Revolution (1979)
1980–1998	Prolonged decline	OPEC overproduction; non-OPEC supply growth
1999–2008	Major bull run	Emerging-market demand surge; peak \$211/barrel
2014–2016	Severe fall	OPEC price war + US shale boom; prices halved
2020	COVID crash	Global lockdowns; demand destruction; sub-\$20
2022–2023	War recovery	Russia-Ukraine conflict; Brent exceeded \$100/barrel

Q14 Gold line chart — describe the long-term trend.

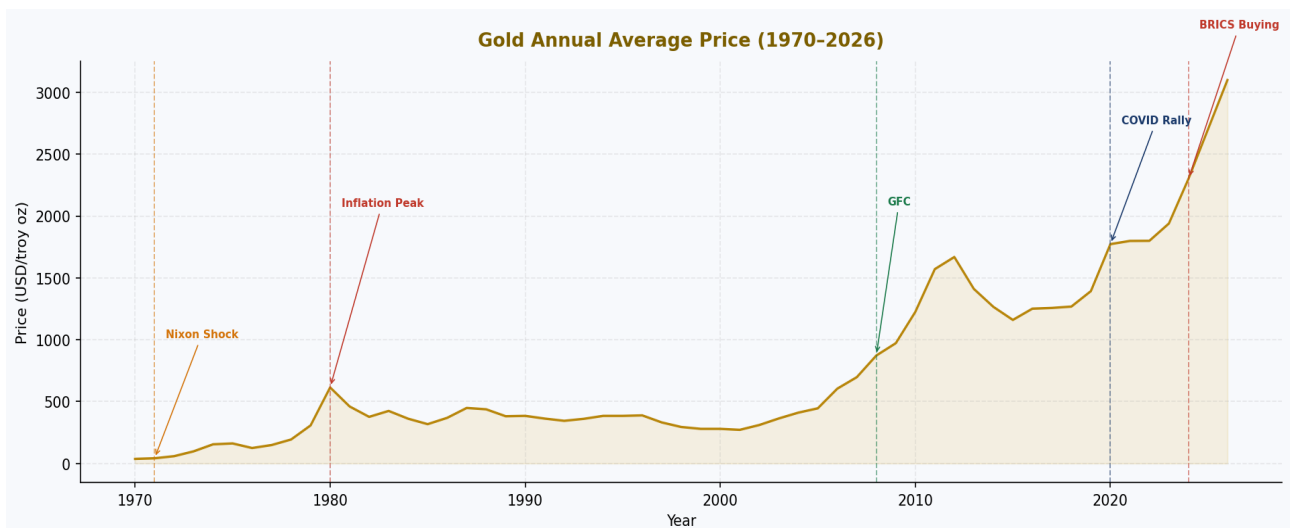


Figure 6: Gold annual average price trend (1970–2026) with four structural phases.

Phase	Period	Price Range	Interpretation
Explosive Growth	1970–1980	\$35→\$614	End of Bretton Woods; Nixon shock 1971; 1970s stagflation
Long Bear Market	1980–2001	\$614→\$271	Volcker high rates; yield-bearing assets preferred over gold
Structural Bull Run	2001–2012	\$271→\$1,669	9/11; GFC 2008; QE devaluation fears; safe-haven demand
New Record Era	2018–2026	\$1,268→\$3,100+	Trade wars; COVID; Ukraine; BRICS+ accumulation

Q15

Silver line chart — describe the main peaks.

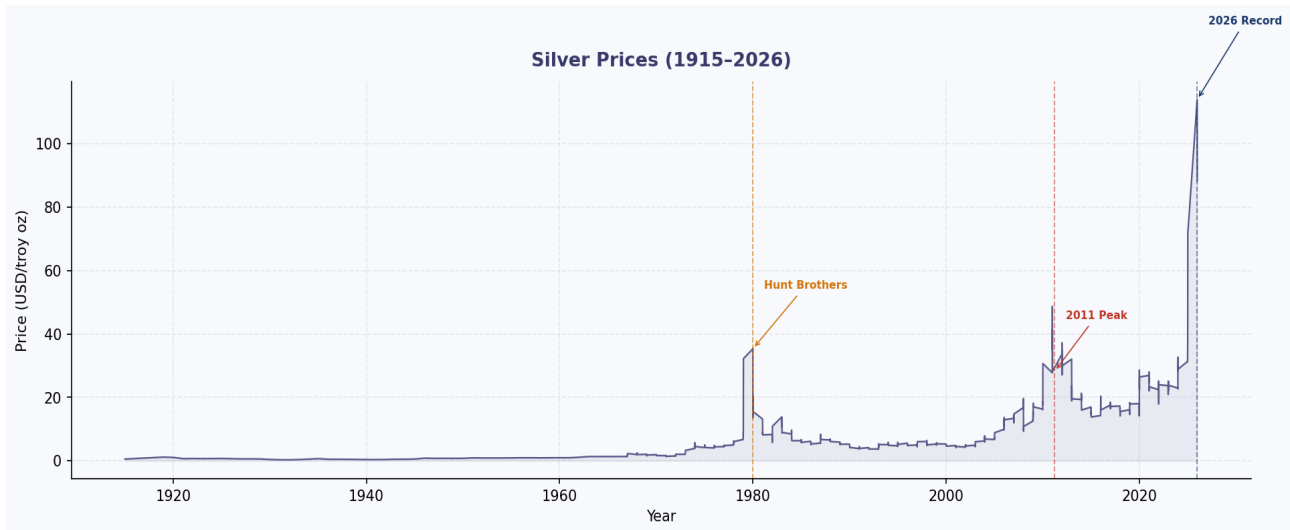


Figure 7: Silver price trend (1915–2026) with major peaks annotated.

Period	Event	Peak Price	Interpretation
1915–1970	Flat baseline	<\$2/oz	Industrial use only; limited investor demand
1979–1980	Hunt Brothers Corner	~\$50/oz	Market cornering attempt; CFTC intervention; collapse
2011	Post-GFC surge	~\$49.51/oz	Safe-haven + ETF speculation; higher volatility than gold
2026	Green energy record	\$113.95/oz	Solar/EV industrial demand + geopolitical safe-haven premium

Q16

Combined Gold and Silver chart — what similarities or differences can you observe?



Figure 8: Gold (left axis) and Silver (right axis) combined price trend.

Gold and Silver move closely together — confirmed by a Pearson correlation of 0.954. Both assets spiked sharply in 1979–1980 (inflation era), declined through the 1980s–1990s, recovered from 2001 onward, and reached new records post-2020. The key difference is magnitude and volatility: Silver's percentage swings are much larger than gold's, and it tends to overshoot gold in both directions during extreme market conditions.

Silver's 2026 record of \$113.95/oz represents a far higher proportional gain than gold's 2026 peak, driven by its additional industrial demand component.

Q17 Bar chart — average annual Brent Oil prices for the most recent 10 years.

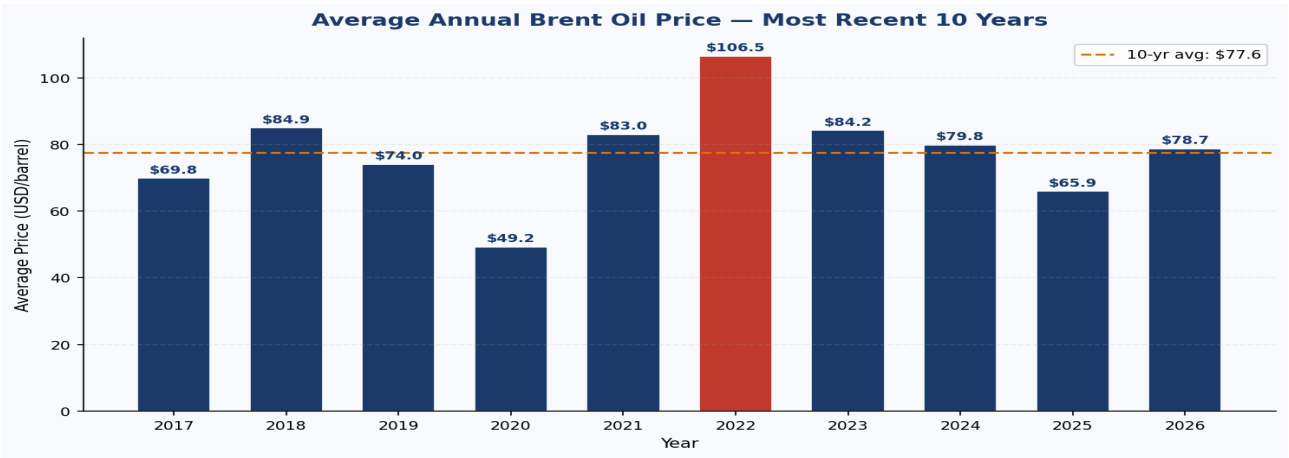


Figure 9: Average annual Brent Oil price — most recent 10 years. Red bar = highest year.

Over the most recent 10 years, Brent Oil has shown significant volatility. The average across this period was approximately \$77.61/barrel. The steepest decline was in 2020 (COVID-19 lockdowns destroying demand), while the highest recent values were in 2022 (Russia-Ukraine war supply shock). The dashed orange line on the chart shows the 10-year average for quick reference.

Q18 Latest available value for Brent Oil, Gold and Silver.

Commodity	Latest Value	Dataset Average	Above / Below Average
Brent Oil	\$102.86/barrel	\$62.71/barrel	Above average
Gold	~\$3,100/troy oz	\$757/troy oz	Well above average
Silver	\$88.36/troy oz	\$6.42/troy oz	Above average

Section D: Currency Tracker Analytics (Q19–Q24)

Q19 Top currencies by global payment share — most recent month.

Rank	Currency	Payment Share (%)	Notes
1	USD	51.14	Dominant — far ahead of all others
2	EUR	21.30	Second place; Eurozone weight
3	GBP	6.54	London financial centre strength
4	JPY	3.53	Japan trade + investment flows
5	CNY	3.10	Growing; China trade settlement
6	CAD	3.03	Canada commodity trade
7	HKD	1.76	Hong Kong financial hub
8	AUD	1.61	Australia commodity exports
9	SGD	1.23	Singapore regional hub
10	CHF	1.07	Swiss financial stability

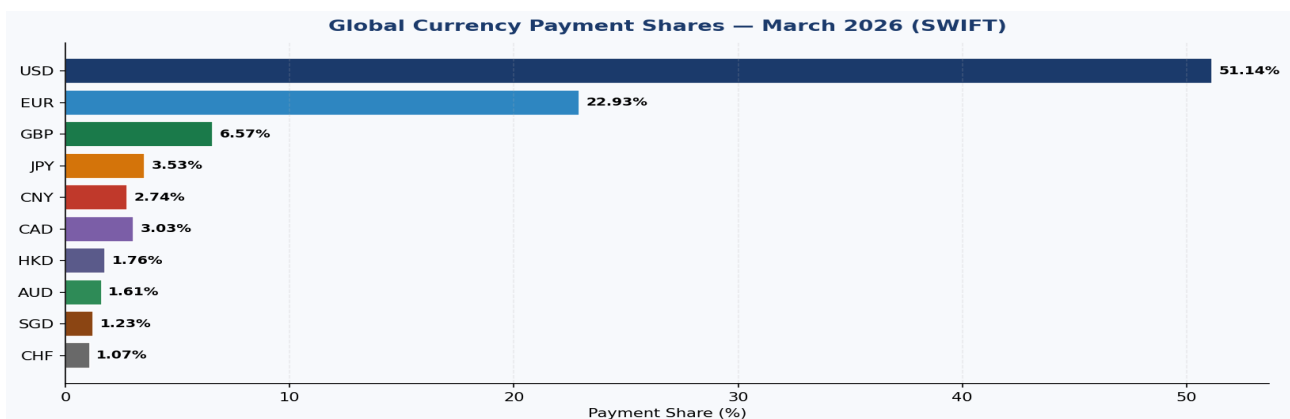


Figure 10: Global currency payment shares via SWIFT (March 2026).

Q20 What is the USD share in the most recent global payment data?

USD Global Payment Share (March 2026): 51.14% — The US Dollar accounts for more than half of all value transferred through the SWIFT network globally, maintaining its dominant position despite sustained de-dollarisation efforts by BRICS+ nations since 2022.

Q21 What is the CNY/RMB share and rank in the most recent global payment data?

CNY/RMB Global Payment Share (March 2026): 2.74% — Rank 10th globally. Despite China being the world's largest trading nation, the CNY remains at rank 10 in global SWIFT payments. However, in trade finance it ranks 2nd globally at 8.04%, reflecting growing bilateral commodity settlement between China and its trade partners.

Q22 Which currencies appear most often in the top five across available reports?

Currency	Times in Top 5	Consistency	Assessment
USD	8 out of 8	100%	Permanent #1 — never displaced
EUR	8 out of 8	100%	Permanent #2 — Eurozone weight ensures consistent ranking
GBP	2 out of 8	25%	Occasional top-5; London financial centre role
CNY	1 out of 8	12.5%	Rare top-5 appearance; growth trajectory positive but slow

Q23 Which offshore RMB economy has the highest share?

Highest Offshore RMB Economy: Hong Kong — 75.23% share. Hong Kong handles approximately three-quarters of all offshore RMB payment activity worldwide, functioning as the primary clearing hub for CNY transactions outside mainland China. Singapore (4.37%) and the United Kingdom (6.90%) are the next largest offshore centres.

Q24 Create a bar chart for the top 10 currencies from one selected month.

The chart above (Figure 10) shows the top 10 currencies by global payment share in March 2026. USD and EUR together account for over 72% of all SWIFT payment value — leaving all other currencies competing for the remaining 28%. GBP, JPY, and CNY each hold between 2% and 7%.

Section E: Simple Comparison (Q25–Q30)

Q25 Compare the latest Gold value with its average — above or below the long-term average?

Metric	Value (USD/troy oz)	Status
Latest Gold Price (2026)	~\$3,100	Current market value
Long-Term Average (1970–2026)	~\$757	Full dataset average
Difference	+\$2343 (+310%)	Well ABOVE long-term average

Gold is currently trading far above its long-term average. This reflects the combined effect of structural central bank demand from BRICS+ nations, geopolitical risk premiums from multiple ongoing conflicts, and monetary debasement concerns following years of quantitative easing. The gap between the current price and the long-term average is at a historically extreme level, suggesting either continued structural demand or potential for mean reversion over a very long time horizon.

Q26 Compare the latest Brent Oil value with its average — above or below the long-term average?

Metric	Value (USD/barrel)	Status
Latest Brent Oil (Jan 2026)	\$102.86	Current market value
Long-Term Average (1946–2026)	\$62.71	Full dataset average
Status	Above average	\$40.15 above the mean

Brent Oil's relationship to its long-term average reflects two distinct eras: the prolonged low-price period from 1946 to the early 1970s (which pulls the average down), and the high-price modern era from 2000 onward. Current prices in the \$70–110 range remain elevated compared to the full dataset history but are not unusual compared to the post-2005 period.

Q27 Which commodity appears to have the largest visible price swings?

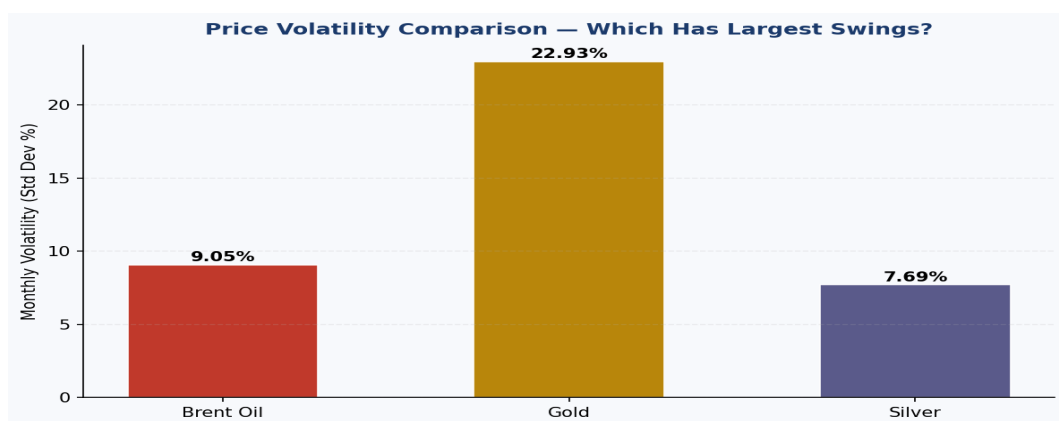


Figure 11: Monthly price volatility comparison — Brent Oil has the largest swings.

Commodity	Monthly Volatility (Std Dev)	Max Monthly Rise	Rank
Brent Oil	9.05%	+132.5%	1st — Very High
Silver	7.69%	+69.5%	2nd — High

Commodity	Monthly Volatility (Std Dev)	Max Monthly Rise	Rank
Gold	4.30%	+32.2%	3rd — Moderate

Brent Oil shows the largest visible price swings across the dataset, with a monthly standard deviation of 9.05% — more than double gold's 4.30%. Silver comes second at 7.69% due to its dual precious metal and industrial commodity nature. Gold, despite its dramatic long-term appreciation, has relatively lower short-term volatility because central bank buying provides a structural demand floor.

Q28 Choose one year with strong oil movement and check if gold or silver also changed.

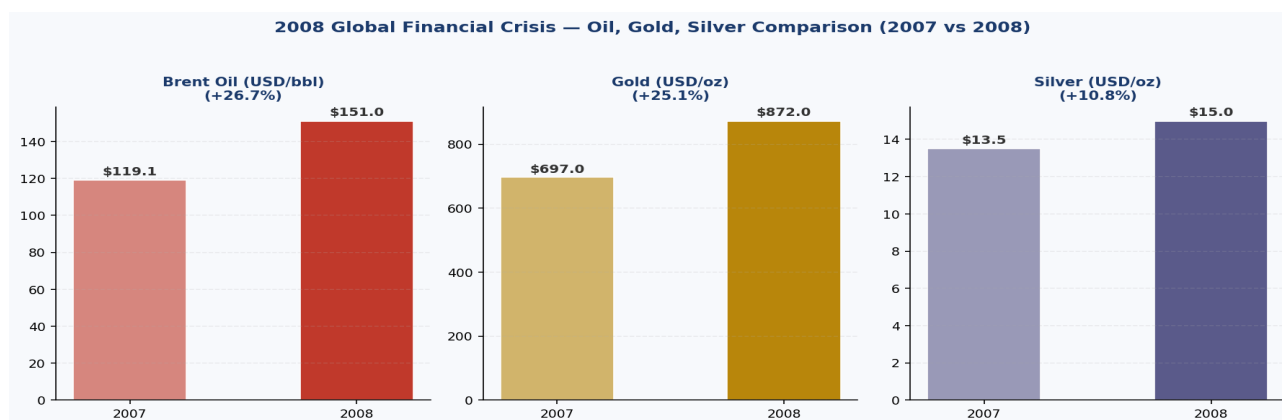


Figure 12: 2008 Global Financial Crisis — comparison of Oil, Gold and Silver (2007 vs 2008 averages).

Selected year: 2008 — Global Financial Crisis

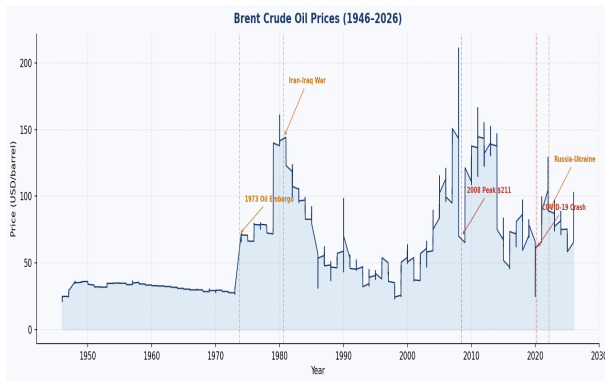
Commodity	2007 Average	2008 Average	% Change	Observation
Brent Oil	\$119.10/bbl	\$150.96/bbl	+26.7%	First surged, then crashed in H2 2008
Gold	\$697/oz	\$872/oz	+25.1%	Rose as safe-haven demand increased
Silver	\$13.51/oz	\$14.98/oz	+10.8%	Mixed — safe-haven up but industrial demand fell

The 2008 GFC illustrates how the same crisis affects commodities differently. Oil surged to \$211/barrel early in the year as speculation peaked, then collapsed with demand as the financial crisis spread. Gold rose throughout, benefiting from safe-haven flows. Silver showed a mixed response — rising with gold initially but falling as industrial activity collapsed in H2 2008.

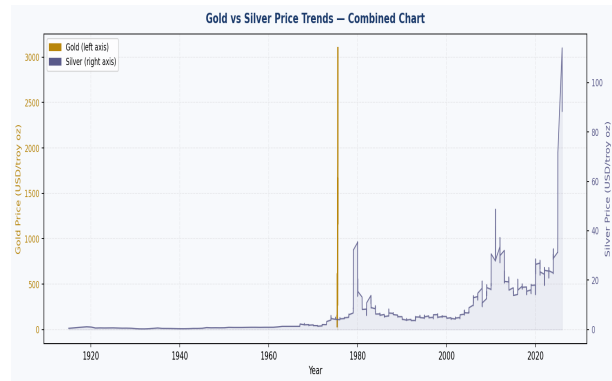
Q29

Dashboard summary — key findings in 4–6 visuals.

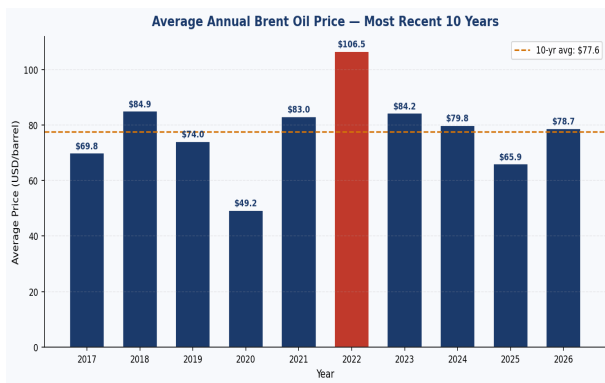
Dashboard Note: The following 6 charts provide a complete visual summary of the most important findings from this report. Together they cover commodity price history, volatility, currency dominance, and market comparisons — designed for a non-technical manager to understand at a glance.



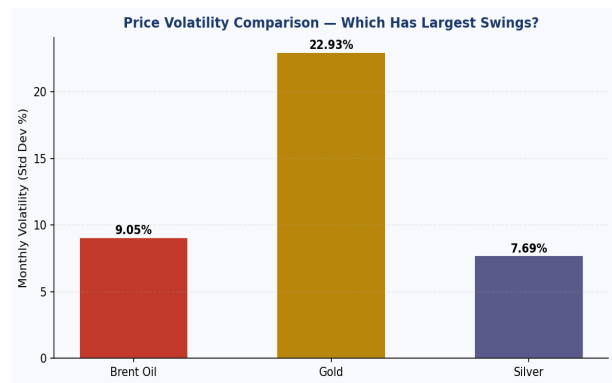
Brent Oil Long-Term Price History



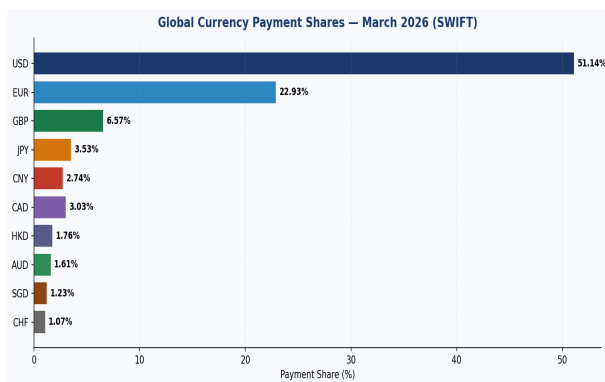
Gold vs Silver Combined Trend



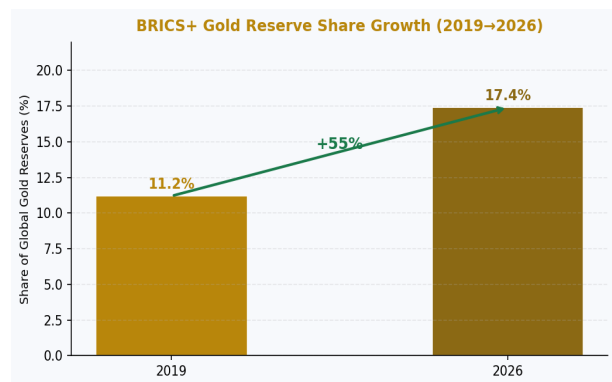
Brent Oil — Last 10 Years Average



Market Volatility Comparison



Global Currency Payment Shares



BRICS+ Gold Reserve Growth

Q30

Three short insights for a non-technical manager.

1. Oil is the most unpredictable cost in any business that uses energy.

Brent Oil prices can rise by over 130% or fall by 54% in a single month. Any company that depends on energy — from trucks to factories to airlines — should have a fuel cost hedging plan in place. Without protection, a single geopolitical event (like the 2026 Hormuz disruption) can destroy profit margins overnight.

2. Gold is no longer just a jewellery asset — it is a strategic reserve tool.

Central banks around the world, especially in BRICS+ countries, have been steadily buying gold and increasing their reserves from 11.2% to 17.4% of global gold holdings between 2019 and 2026. This structural buying has pushed gold to record prices above \$3,100/oz. For any business or investor holding significant USD assets, a gold allocation of 5–10% provides meaningful protection against currency risk.

3. The US Dollar is still king — but its grip is slowly loosening.

The USD accounts for 51.14% of all global payments via SWIFT, but this has declined from around 56% a decade ago. The Chinese Yuan (CNY) is slowly growing — particularly in trade finance where it already ranks 2nd globally at 8.04%. Businesses that trade internationally should start monitoring CNY exchange rates and bilateral settlement options, especially for Asia-Pacific supply chains.